A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment

Introduction and Research Objective

Ersoy and Erşahin begin with a familiar weakness of large language models. Their knowledge sits in their parameters, so it can go out of date, and the model sometimes invents specific details. In finance and e-commerce, where a wrong figure or an outdated policy has real consequences, this is a serious problem. Retrieval-Augmented Generation (RAG) reduces it by supplying retrieved passages at inference time. The authors argue that the four main retrieval styles (sparse, dense, hybrid and fusion) are hard to compare fairly in earlier work, because those studies mix retrieval quality with generation quality, leave key hyperparameters unspecified, and report single scores without uncertainty. The paper therefore aims at a reproducible and statistically careful comparison, organised around five research questions on comparative performance, sensitivity to tuning, latency–utility trade-offs, auditability, and robustness across two hosting setups. The authors state that the contribution is methodological and deployment-oriented rather than a new retrieval algorithm.

Methodology and Architectures

The authors treat RAG as a spectrum. Sparse retrieval uses BM25 over an inverted index. Its scoring is fully inspectable and fast, but it relies on exact word overlap and so struggles with paraphrased or multilingual queries. Dense retrieval embeds queries and documents as vectors and searches them with approximate nearest-neighbour methods (HNSW with a Sentence-Transformer encoder in their configuration). It finds passages that share meaning but not wording, at the price of costly indexing and similarity scores that are hard to explain. Hybrid retrieval runs BM25 and dense search in parallel, merges the two ranked lists with reciprocal-rank fusion, and applies a cross-encoder re-ranker. Fusion retrieval is the most elaborate: a language model expands the query into several sub-queries (four in their setup), the results are combined, and in the Fusion-in-Decoder variant the decoder weighs passages while generating. The paper describes the strengths and weaknesses of each design with worked examples, and each is implemented as a parameterised LangChain template.

To isolate the effect of retrieval, the generator is held fixed within each study cell. A zero-shot baseline uses the same generator and prompt with an empty context block. Two generators were used: cloud-hosted GPT-4o and an on-premises Llama-3-8B-Instruct, quantised and run through llama.cpp on a MacBook Pro. Both used temperature 0.2, top-p 0.9 and a 512-token output limit. Table 2 of the paper lists the per-pipeline settings, such as chunk sizes between 256 and 512 tokens, initial top-k values and re-ranking depth.

Experimental Setup and Evaluation

Two corpora were built from public sources. The finance corpus has 3,214 documents (about 92.7 million tokens) drawn from SEC filings, investor relations material and central bank circulars from June 2019 to June 2025. The e-commerce corpus has 4,836 passages (about 68.3 million tokens) from product pages and policy documents in English and Turkish. A separate evaluation evidence pool contains 7,150 passages. The test set has 500 questions stratified by domain and difficulty, of which 75% are answerable and 25% deliberately unanswerable, so that a system cannot look good

simply by abstaining. Two annotators labelled a 20% sample, with Cohen's kappa of 0.77 for answerability, 0.69 for domain and 0.74 for difficulty.

Systems were scored with the RAGAS protocol on faithfulness, answer relevance, context precision and context recall, and the authors added latency and an “audibility” measure, defined as the share of answer claims backed by a valid citation that meets a recency threshold. Results are reported as means with 95% bootstrap confidence intervals, and pipelines were compared with paired Wilcoxon tests using Holm correction.

Key Findings

Every retrieval-grounded pipeline clearly beat the zero-shot baseline. GPT-4o without retrieval produced wrong monetary figures or outdated regulatory thresholds in roughly one-third of cases, while the RAG pipelines brought such errors down to single-digit percentages. In Table 3, the baseline's faithfulness is 0.58 and its context recall 0.38. Sparse retrieval reaches 0.70 faithfulness and 0.71 audibility, but its recall is only 0.62 and its latency is 880 ms against 680 ms for the baseline. Dense retrieval has the highest recall (0.78) and 0.74 faithfulness, at 1,020 ms. Hybrid retrieval reaches 0.80 faithfulness, 0.83 answer relevance and 0.79 audibility at 1,110 ms. Fusion scores highest on faithfulness (0.82) and answer relevance (0.84), with 0.71 context precision, but it is also the slowest at 1,230 ms.

The authors summarise this as a trade-off: fusion gives the strongest factual fidelity, hybrid gives the most balanced profile with moderate latency, sparse is best when traceability is the priority, and dense offers the widest semantic reach. They also note that answers citing primary and recent sources, such as EDGAR filings or official policy pages, were consistently more faithful. One point deserves caution. The prose says sparse retrieval had the highest context precision and fusion reduced auditability, yet Table 3 lists fusion with the highest precision (0.71) and auditability (0.81). I could not reconcile these statements from the paper alone, so the table values should be treated as the more precise evidence.

Conclusion and Critical Understanding

The authors conclude that all four pipelines improve factual behaviour on this workload and that the right choice depends on the deployment. Sparse or hybrid suits settings dominated by auditability and strict latency, dense or hybrid suits multilingual or paraphrase-heavy traffic, and fusion suits cases where factual coverage matters most and batch or offline operation is acceptable, ideally with citation filtering.

They acknowledge several limits. The evaluation covers only finance and e-commerce, and different generators, encoders, ANN indexes or prompt budgets could change the outcomes. RAGAS and similar automatic metrics are still proxies for human judgement, and cost and carbon accounting were out of scope. Future work proposed includes dynamic indexing, retrieval-aware decoding with joint training, long-context generators with claim-level citation alignment, and cross-domain replications with human-in-the-loop judgements. In my own reading, the paper's main value is its transparent protocol, though the full production datasets and code are not released before publication, which limits independent verification.